

# GROWW F&O ALGO TRADING BOT

Complete Operational & Strategy Manual • NIFTY & BANKNIFTY Options Terminal

**EXECUTIVE OVERVIEW:** This document provides the complete end-to-end user guide for the Groww Algorithmic Trading Bot. Specifically engineered for **NSE Futures & Options (F&O)**, the system tracks real-time spot movements on **NIFTY 50** and **BANKNIFTY**, dynamically selects At-The-Money (ATM) strikes, and executes Call (CE) or Put (PE) option orders based on Dual EMA, RSI, and Supertrend indicators. It includes a live Google Chrome web terminal protected with login authentication.

## 1. Starting the Bot & Accessing the Terminal

| Execution Method                         | Command / Action                                               | Description & Behavior                                                                                                                                                                |
|------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Option 1: Chrome Dashboard (Recommended) | Double-click run_chrome_dashboard.bat<br>or run: python app.py | Starts background F&O scanner thread, hosts Flask web server on port 5000, and auto-launches Google Chrome at <a href="http://127.0.0.1:5000/login">http://127.0.0.1:5000/login</a> . |
| Option 2: Terminal Console Runner        | Double-click run_bot.bat<br>or run: python main.py             | Runs terminal-based continuous scanner displaying live index spots, indicators, CE/PE premiums, and simulated lot fills in text mode.                                                 |

## 2. Web Terminal Login & Broker Credentials

| Parameter        | Configured Value                                                      | Operational Notes                                                          |
|------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------|
| Login URL        | <a href="http://127.0.0.1:5000/login">http://127.0.0.1:5000/login</a> | Direct login access; Chrome auto-opens to this page upon startup.          |
| Username         | admin                                                                 | Configurable via DASHBOARD_USERNAME in config.py or .env.                  |
| Password         | groww123                                                              | Configurable via DASHBOARD_PASSWORD in config.py or .env.                  |
| Quick Auto-Login | ■ 1-Click Button on Login Screen                                      | One-click button auto-fills credentials and signs in immediately.          |
| Connected UCC    | 6629599909                                                            | Authenticated Groww Trading Account with active ['CASH', 'FNO'] segments.  |
| Session Logout   | ■ Single-Click Logout in Navbar                                       | Instantly clears encrypted session and redirects back to the login screen. |

### 3. Trading Type & Contract Specifications

The bot executes **NSE Futures & Options (F&O) Intraday Options Buying** on India's primary benchmark indices: **NIFTY 50** and **BANKNIFTY**.

| Underlying Index | Lot Size        | Strike Step                    | Traded Contracts           | Product Code   |
|------------------|-----------------|--------------------------------|----------------------------|----------------|
| NIFTY 50 Index   | 25 Shares / Lot | 50 Points (e.g. 25100, 25150)  | At-The-Money (ATM) CE / PE | MIS (Intraday) |
| BANKNIFTY Index  | 15 Shares / Lot | 100 Points (e.g. 51400, 51500) | At-The-Money (ATM) CE / PE | MIS (Intraday) |

#### Dual Trading Modes:

- **Paper F&O Simulation (Zero Risk):** Simulated with **Rs. 1,00,000.00** virtual funds. Calculates Greek delta (0.50 ATM), premium fluctuations, trailing P&L, stop-loss triggers, and target fills without risking real money.
- **Live Groww F&O Trading:** Direct bridge with Groww API (GrowwAPI.place\_order). Places real market orders on NSE F&O using live margin. Instant switching between Paper and Live takes under 15ms via the dashboard button.

### 4. Entry Strategy: When Does the Bot BUY?

The bot continuously computes three mathematical indicators over 1-minute candlestick data to eliminate false breakouts and confirm institutional momentum:

1. **Dual EMA (9 & 21):** Measures fast short-term trend alignment relative to medium-term baseline.
2. **RSI (Period 14):** Confirms momentum strength without entering in overextended zones.
3. **Supertrend (Period: 10, Multiplier: 3.0):** Volatility-based ATR trailing filter setting trend direction.

| Trade Action                                         | Strategy Condition Matrix                                                                                                                                                                                                 | Executed Option Contract                                                                                                                         |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| ■ <b>BUY CALL OPTION (CE)</b><br>(Bullish Entry)     | <b>All 3 Conditions Must Align:</b><br>1. <b>EMA Bullish:</b> Fast EMA(9) > Slow EMA(21) (or Bullish Crossover)<br>2. <b>RSI Strong:</b> RSI(14) ≥ 48.0 (Upward Momentum)<br>3. <b>Supertrend:</b> GREEN (Direction = +1) | Bot calculates nearest ATM Strike:<br>ATM = round(Spot / Step) * Step<br>Buys <b>At-The-Money Call (CE)</b><br><i>Example: NIFTY 25150 CE</i>    |
| ■ <b>BUY PUT OPTION (PE)</b><br>(Bearish Entry)      | <b>All 3 Conditions Must Align:</b><br>1. <b>EMA Bearish:</b> Fast EMA(9) < Slow EMA(21) (or Bearish Crossover)<br>2. <b>RSI Weak:</b> RSI(14) ≤ 52.0 (Downward Momentum)<br>3. <b>Supertrend:</b> RED (Direction = -1)   | Bot calculates nearest ATM Strike:<br>ATM = round(Spot / Step) * Step<br>Buys <b>At-The-Money Put (PE)</b><br><i>Example: BANKNIFTY 51500 PE</i> |
| ■ <b>NO TRADE (HOLD)</b><br>(Choppy / Consolidation) | Any condition where EMA, RSI, and Supertrend do not agree unanimously. Protects capital during sideways market chop.                                                                                                      | No new orders placed. Existing positions monitored for exits.                                                                                    |

## 5. Exit Strategy & Risk Rules: When Does the Bot SELL?

Strict risk management governs every open contract. The bot never holds losing positions indefinitely and enforces a disciplined **1:2 Risk-to-Reward Ratio**:

| Exit Trigger                                            | Threshold / Rule                                   | Formula & Concrete Example                                                                                                              |
|---------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ■ <b>STOP-LOSS (SL) HIT</b><br>(Capital Protection)     | -20.0% Loss on Option Entry Premium                | $SL = \text{Entry Premium} \times (1 - 0.20)$<br><i>If bought @ Rs. 100.00, SL triggers at Rs. 80.00. Automatic square-off.</i>         |
| ■ <b>TARGET PROFIT (TP)</b><br>(Profit Booking)         | +40.0% Gain on Option Entry Premium                | $\text{Target} = \text{Entry Premium} \times (1 + 0.40)$<br><i>If bought @ Rs. 100.00, TP triggers at Rs. 140.00. Profit locked in.</i> |
| ■ <b>DAILY CIRCUIT BREAKER</b><br>(Disaster Protection) | Rs. -5,000.00 Max Daily Realized Loss              | If cumulative realized loss hits Rs. 5,000 in a trading session, all trading is immediately halted.                                     |
| ■ <b>POSITION LIMITS</b><br>(Exposure Control)          | Max 2 Concurrent Contracts<br>Max 2 Lots per Trade | Prevents over-leveraging; max 50 Qty in NIFTY (2 lots) and 30 Qty in BANKNIFTY (2 lots).                                                |

## 6. Frequently Asked Questions (FAQ) & Operational Notes

### Q: Why does Live Mode show Rs. 0.00 balance when I switch?

A: In Live Mode, the bot queries your real Groww account margin via Groww API. If you have not yet deposited funds into your Groww Stocks / F&O balance, it accurately reports Rs. 0.00. Once you add funds via the Groww App, the live balance updates automatically.

### Q: How fast does the Paper / Live switch button work?

A: Mode switching has been optimized with asynchronous in-memory caching and responds in under 15 milliseconds (< 0.02s) without freezing your browser.

### Q: What if I forget the terminal login password?

A: The default login is 'admin' and 'groww123'. You can also click the '■ Quick Auto-Login' button on the login screen to sign in instantly.

### Q: Does the bot trade automatically without human intervention?

A: Yes. Once started, the scanner loop scans the index every 3 seconds, monitors active option premiums against SL/TP thresholds, and executes orders automatically.

**DISCLAIMER:** Futures & Options trading involves substantial risk of capital loss. This software is provided for algorithmic automation and informational simulation. Always perform thorough paper testing before committing substantial capital to live market trading.